MATH 2R03 - Assignment 2

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Question 1

Let V be a vector space over a field $\mathbb{F}$, and let $v_1, \ldots, v_m$ be a list of vectors in V. For $k = 1, \ldots, m$ set $w_k := v_1 + \ldots + v_k$. Show that $\text{span}(w_1, \ldots, w_m) = \text{span}(v_1, \ldots, v_m)$.

Solution:

To prove that $\text{span}(w_1, \ldots, w_m) = \text{span}(v_1, \ldots, v_m)$, we must show that $\text{span}(w_1, \ldots, w_m) \subseteq \text{span}(v_1, \ldots, v_m)$ and $\text{span}(v_1, \ldots, v_n)$.

(I) Prove $\text{span}(w_1, \ldots, w_m) \subseteq \text{span}(v_1, \ldots, v_m)$.

Let $u \in \text{span}(w_1, \ldots, w_m)$. By definition of span, we know that there exists $\lambda_1, \ldots, \lambda_m \in \mathbb{F}$ such that:

$$\lambda_1 w_1 + \ldots + \lambda_m w_m = u$$

However, we know that $w_k = v_1 + \ldots + v_k$ by its definition. Therefore, $\lambda_k w_k = \lambda_k(v_1 + \ldots + v_k) = \lambda_k v_1 + \ldots + \lambda_k v_k$. Therefore, we have that $\lambda_k w_k + \lambda_{k+1} w_{k+1} + \ldots + \lambda_m w_m = (\lambda_k + \lambda_{k+1} + \ldots \lambda_m) v_k + (\lambda_{k+1} + \ldots + \lambda_m) v_{k+1} + \ldots \lambda_m v_m$ and hence:

$$\begin{aligned}
\lambda_1 v_1 + \ldots + \lambda_m(v_1 + \ldots + v_m) = u &\implies \\
\lambda_1 v_1 + \ldots + \lambda_m v_1 + \ldots + \lambda_m v_m = u &\implies \\
(\lambda_1 + \ldots + \lambda_m) v_1 + \ldots + \lambda_m v_m = u &
\end{aligned}$$

Thus, since $(\lambda_k + \ldots + \lambda_m) \in \mathbb{F}$, letting $\alpha_k \in \mathbb{F}$ such that $\alpha_k = \sum_{i=k}^{m} \lambda_i$, we have that:

$$\alpha_1 v_1 + \ldots + \alpha_m v_m = u$$

And thus $u \in \text{span}(v_1, \ldots, v_m)$. Therefore, $\text{span}(w_1, \ldots w_m) \subseteq \text{span}(v_1, \ldots, v_m)$.

(II) Prove $\text{span}(v_1, \ldots, v_k) \subseteq \text{span}(w_1, \ldots, w_k)$.

Let $u \in \text{span}(v_1, \ldots v_m)$. By definition of span, we know that there exists $\lambda_1, \ldots, \lambda_m \in \mathbb{F}$ such that:

$$\lambda_1 v_1 + \ldots + \lambda_m v_m = u$$

Or in other words:

$$\sum_{k=1}^{m} \lambda_k v_k = u$$

Now, let $\alpha_k \in \mathbb{F}$ such that $\alpha_m \coloneqq \lambda_m$ and $\alpha_k \coloneqq \lambda_k - \lambda_{k+1}$ for all $1 \leq k < m$. (Thus, $\alpha_m = \lambda_m, \alpha_{m-1} = \lambda_{m-1} - \lambda_m, \ldots, \alpha_1 = \lambda_1 - \lambda_2$). Now, we know that, for all $1 \leq k < m$:

$$\begin{aligned}
\alpha_k w_k &= \alpha_k(v_1 + \ldots + v_k) \\
&= (\lambda_k - \lambda_{k+1})(v_1 + \ldots + v_k) \\
&= (\lambda_k - \lambda_{k+1}) \sum_{i=1}^{k} (v_i)
\end{aligned}$$

This therefore means that:

$$\begin{aligned}
\alpha_1 w_1 + \ldots + \alpha_m w_m &= \sum_{k=1}^{m-1} \alpha_k w_k + \alpha_m w_m \\
&= \sum_{k=1}^{m-1} \left((\lambda_k - \lambda_{k+1}) \left(\sum_{i=1}^{k} (v_i) \right) \right) + \alpha_m w_m \\
&= \sum_{k=1}^{m-1} \left(\lambda_k \sum_{i=1}^{k} (v_i) - \lambda_{k+1} \sum_{i=1}^{k} (v_i) \right) + \alpha_m w_m \\
&= \sum_{k=1}^{m-1} \left(\lambda_k \sum_{i=1}^{k} (v_i) \right) - \sum_{k=1}^{m-1} \left(\lambda_{k-1} \sum_{i=1}^{k} (v_i) \right) + \alpha_m w_m \\
&= \sum_{k=1}^{m-1} \left(\lambda_k \sum_{i=1}^{k} (v_i) \right) - \sum_{k=2}^{m} \left(\lambda_k \sum_{i=1}^{k-1} (v_i) \right) + \alpha_m w_m \\
&= \lambda_1 v_1 + \sum_{k=2}^{m-1} \left(\lambda_k \sum_{i=1}^{k} (v_i) \right) - \sum_{k=2}^{m} \left(\lambda_k \sum_{i=1}^{k-1} (v_i) \right) + \alpha_m w_m \\
&= \lambda_1 v_1 + \sum_{k=1}^{m-1} \left(\lambda_k \sum_{i=1}^{k} (v_i) \right) - \sum_{k=2}^{m-1} \left(\lambda_k \sum_{i=1}^{k-1} (v_i) \right) - \lambda_m \sum_{i=1}^{m-1} (v_i) + \alpha_m w_m \\
&= \lambda_1 v_1 + \sum_{k=2}^{m-1} \left(\lambda_k \sum_{i=1}^{k} (v_i) - \lambda_k \sum_{i=1}^{k-1} (v_i) \right) - \lambda_m \sum_{k=1}^{m-1} (v_i) + \alpha_m w_m \\
&= \lambda_1 v_1 + \sum_{k=2}^{m-1} \left(\lambda_k v_k + \lambda_k \sum_{i=1}^{k-1} (v_i) - \lambda_k \sum_{i=1}^{k-1} (v_i) \right) - \lambda_m \sum_{k=1}^{m-1} (v_i) + \alpha_m w_m \\
&= \lambda_1 v_1 + \sum_{k=2}^{m-1} (\lambda_k v_k) - \lambda_m \sum_{k=1}^{m-1} (v_i) + \alpha_m w_m \\
&= \sum_{k=1}^{m-1} (\lambda_k v_k) - \lambda_m \sum_{k=1}^{m-1} (v_i) + \lambda_m \sum_{i=1}^{m} (v_i) \\
&= \sum_{k=1}^{m-1} (\lambda_k v_k) - \lambda_m \sum_{k=1}^{m-1} (v_i) + \lambda_m \sum_{i=1}^{m-1} (v_i) + \lambda_m v_m \\
&= \sum_{k=1}^{m-1} (\lambda_k v_k) + \lambda_m v_m \\
&= \sum_{k=1}^{m} (\lambda_k v_k)
\end{aligned}$$

And since $\sum_{k=1}^{m} (\lambda_k v_k) = u$ as stated prior, we have that:

$$\alpha_1 w_1 + \ldots + \alpha_m w_m = u$$

And thus $u \in \text{span}(w_1, \ldots, w_m)$. Therefore, $\text{span}(v_1, \ldots, v_m) \subseteq \text{span}(w_1, \ldots, w_2)$.

Therefore, since $\text{span}(w_1, \ldots, w_m) \subseteq \text{span}(v_1, \ldots, v_m)$ and $\text{span}(v_1, \ldots, v_m) \subseteq \text{span}(w_1, \ldots, w_m)$, $\text{span}(v_1, \ldots, v_m) = \text{span}(w_1, \ldots, w_m)$.

Question 2

Prove or give a counterexample: $v_1, \ldots, v_m$ is a linearly independent list of vectors in V, then $5v_1 - 4v_2, v_2, \ldots, v_m$ is linearly independent.

Solution:

Let $\lambda_1, \ldots, \lambda_m \in \mathbb{F}$. We can see that:

$$\begin{gathered}
\lambda_1(5v_1 - 4v_2) + \lambda_2 v_2 + \ldots + \lambda_m v_m = 0 \implies \\
5\lambda_1 v_1 - 4\lambda_1 v_2 + \lambda_2 v_2 + \ldots + \lambda_m v_m = 0 \implies \\
5\lambda_1 v_1 + (\lambda_2 - 4\lambda_1)v_2 + \ldots + \lambda_m v_m = 0
\end{gathered}$$

Now since $v_1, \ldots, v_m$ are linearly independent, we know that this implies $5\lambda_1 = (\lambda_2 - 4\lambda_1) = \ldots = \lambda_m = 0$. Thus, if $5\lambda_1 = 0$, then $\lambda_1 = 0$, and if $\lambda_1 = 0$, then $\lambda_2 - 4\lambda_1 = 0$ implies $\lambda_2 = 0$. Thus, we have that:

$$\lambda_1(5v_1 - 4v_2) + \lambda_2 v_2 + \ldots + \lambda_m v_m = 0$$

Implies $\lambda_1 = \ldots = \lambda_m = 0$ and hence $5v_1 - 4v_2, v_2 \ldots, v_m$ is linearly independent.

Question 3

Let $v_1, \ldots, v_m$ be a linearly independent list of vectors in V, and let $w \in V$. Show that if the list $v_1, \ldots, v_m, w$ is a linearly independent list if and only if $w \notin \text{span}(v_1, \ldots, v_m)$.

Solution:

We will prove this by showing the forward and reverse directions separately.

(I) Forward direction.

Assume, for the purpose of contradiction, that $v_1, \ldots, v_m, w$ is a linearly independent list but that $w \in \text{span}(v_1, \ldots, v_m)$. Then by definition of span, we know there exists $\lambda_1, \ldots, \lambda_m \in \mathbb{F}$ such that:

$$\lambda_1 v_1 + \ldots + \lambda_m v_m = w$$

However, note that this would imply that:

$$\lambda_1 v_1 + \ldots + \lambda_m v_m + (-w) = 0$$

Which implies that $v_1, \ldots, v_m, w$ cannot be a linearly independent list of vectors since w has a non-zero coefficient. Therefore, by contradiction, $v_1, \ldots, v_m, w$ being a linearly independent list implies that $w \notin \text{span}(v_1, \ldots, v_m)$.

(II) Reverse direction

Once again, assume for the purposes of contradiction that $w \notin \text{span}(v_1, \ldots, v_m)$ but that both $v_1, \ldots, v_m$ and $v_1, \ldots, v_m, w$ are linearly dependent lists of vectors. Then, we know there exists $\lambda_1, \ldots, \lambda_m, \lambda_{m+1} \in \mathbb{F}$ where some $\lambda_k \neq 0$ such that:

$$\lambda_1 v_1 + \ldots + \lambda_m v_m + \lambda_{m+1} w = 0$$

However, this would imply that:

$$\lambda_1 v_1 + \ldots + \lambda_m v_m = -\lambda_{m+1} w$$

Now, if $\lambda_{m+1} \neq 0$, then we have that:

$$-\frac{\lambda_1}{\lambda_{m+1}}v_1 + \ldots - \frac{\lambda_m}{\lambda_{m+1}}v_m = w$$

And thus, $w \in \text{span}(v_1, \ldots, v_m)$ which would be a contradiction. If $\lambda_{m+1} = 0$, then we would have that:

$$\begin{aligned}&\lambda_1 v_1 + \ldots + \lambda_m v_m = -\lambda_{m+1} w \implies \\ &\lambda_1 v_1 + \ldots + \lambda_m v_m = 0\end{aligned}$$

And since at least one $\lambda_k \neq 0$, that would imply that $v_1, \ldots, v_m$ is not a linearly independent list of vectors, which would also be a contradiction. Therefore, $v_1, \ldots, v_m$ being an independent list of vectors and $w \notin \text{span}(v_1, \ldots, v_m)$ implies that $v_1, \ldots, v_m, w$ is a linearly independent list of vectors.

Thus, for a linearly independent list of vectors $v_1, \ldots, v_m$, we have that $v_1, \ldots, v_m, w$ is also a linearly independent list of vectors if and only if $w \notin \text{span}(v_1, \ldots, v_m)$.

Question 4

Let $\mathbb{F}$ be a field. For $n \in \mathbb{N}$, let:

$$\mathbb{F}[x]_n \coloneqq \{\, p \in \mathbb{F}[x] \colon \deg(p) \leq n \,\}$$

(a) Prove that $\mathbb{F}[x]_n$ is a subspace of $\mathbb{F}[x]$.

Solution:

By definition of subspaces, for $\mathbb{F}[x]_n$ to be a subspace of $\mathbb{F}[x]$, then $0 \in \mathbb{F}[x] \cap \mathbb{F}[x]_n$, $a, b \in \mathbb{F}[x]_n \implies a + b \in \mathbb{F}[x]_n$, and $a \in \mathbb{F}[x]_n \implies \forall \lambda \in \mathbb{F}, \lambda a \in \mathbb{F}[x]_n$. We will prove each of these claims separately.

(I) Prove $0 \in \mathbb{F}[x] \cap \mathbb{F}[x]_n$.

Since 0 is polynomial and for any polynomial p, $p + 0 = p$, we know that 0 must be the additive identity of $\mathbb{F}[x]$ and hence $0 \in \mathbb{F}[x]$. The degree of the 0 polynomial is evidently 0, and since $n \in \mathbb{N} \implies n \geq 0$, $0 \in \mathbb{F}[x]_n$.

(II) Prove $a, b \in \mathbb{F}[x]_n \implies (a + b) \in \mathbb{F}[x]_n$.

Let $\alpha, \beta = \deg(a), \deg(b)$ respectively. By the construction of $\mathbb{F}[x]_n$, we know that $\alpha, \beta \leq n$. By lemma 2.2 presented in lecture 6, we know that $\deg(a + b) \leq \max\{\, \alpha, \beta \,\}$. But since $\alpha, \beta \leq n$, we have that $\deg(a + b) \leq \max \alpha, \beta \leq \max n, n = n$. Thus, $a + b \in \mathbb{F}[x]_n$.

(III) Prove $a \in \mathbb{F}[x]_n \implies \forall \lambda \in \mathbb{F}, \lambda a \in \mathbb{F}[x]_n$.

Let $\alpha = \deg(p)$ and let $\lambda \in \mathbb{F}$ be arbitrary. By the construction of $\mathbb{F}[x]_n$, we know that $\alpha \leq n$. By lemma 2.1 presented in lecture 6, we know that $\deg(\lambda a) \leq \deg(a) = \alpha$, however since $\alpha \leq n$, we have that $\deg(\lambda a) \leq \deg(a) = \alpha \leq n$. Thus, $\lambda a \in \mathbb{F}[x]_n$.
Therefore, $\mathbb{F}[x]_n$ is a subspace of $\mathbb{F}[x]$.

(b) Explain why there does not exist a list of six polynomials that is linearly independent in $\mathbb{F}[x]_4$

Solution:

We know that the standard basis of $\mathbb{F}[x]_4$ is given by $(1,\ x,\ x^2,\ x^3,\ x^4)$ which is a list of length five. Since, for any vector space, any list with more vectors than any of its bases cannot be linearly independent, we therefore know that any list of six polynomials cannot be linearly independent in $\mathbb{F}[x]_4$

(c) Explain why no list of four polynomials spans $\mathbb{F}[x]_4$.

Solution:

We know that the standard basis of $\mathbb{F}[x]_4$ is given by $(1,\ x,\ x^2,\ x^3,\ x^4)$ which is a list of length five. Since, for any vector space, any list with fewer vectors than any of its bases cannot span the space, we therefore know that any list of four polynomials cannot span $\mathbb{F}[x]_4$.

Question 5

Consider the set $F_3 \coloneqq \{0, 1, 2\}$ together with the addition $a \oplus b \coloneqq (a + b) \mod 3$ and the multiplication $a \otimes b \coloneqq (ab) \mod 3$.

(a) Verify that $\oplus$ and $\otimes$ satisfy the field axioms; only submit here the proof of distributivity.

Solution:

Let $a, b, c \in F_3$. To prove that $\oplus$ and $\otimes$ are distributive. We must show that $a \otimes (b \oplus c) = a \otimes b \oplus a \otimes c$.

$$\begin{aligned}
a \otimes (b \oplus c) &= a \otimes ((b + c) \mod 3) && \text{By definition of } \oplus \\
&= a((b + c) \mod 3) \mod 3 && \text{By definition of } \otimes \\
&= ((a(b \mod 3)) + (a(c \mod 3))) \mod 3 && \text{By distributivity of modulo} \\
&= (((ab) \mod 3) + ((ac) \mod 3)) \mod 3 && \text{By property of modulo over multiplication} \\
&= ((a \otimes b) + (a \otimes c)) \mod 3 && \text{By definition of } \otimes \\
&= (a \otimes b) \oplus (a \otimes c) && \text{By definition of } \oplus
\end{aligned}$$

(b) List all functions in the vector space ${F_3}^{F_3}$ (how many are there?), and determine which of these are polynomial functions.

Solution:

There are a total of 27 functions in the vector space ${F_3}^{F_3}$:

$$f(2)=0 \quad f(1)=0 \quad f(0)=0 \tag{1}$$
$$f(2)=0 \quad f(1)=0 \quad f(0)=1 \tag{2}$$
$$f(2)=0 \quad f(1)=0 \quad f(0)=2 \tag{3}$$
$$f(2)=0 \quad f(1)=1 \quad f(0)=0 \tag{4}$$
$$f(2)=0 \quad f(1)=1 \quad f(0)=1 \tag{5}$$
$$f(2)=0 \quad f(1)=1 \quad f(0)=2 \tag{6}$$
$$f(2)=0 \quad f(1)=2 \quad f(0)=0 \tag{7}$$
$$f(2)=0 \quad f(1)=2 \quad f(0)=1 \tag{8}$$
$$f(2)=0 \quad f(1)=2 \quad f(0)=2 \tag{9}$$
$$f(2)=1 \quad f(1)=0 \quad f(0)=0 \tag{10}$$
$$f(2)=1 \quad f(1)=0 \quad f(0)=1 \tag{11}$$
$$f(2)=1 \quad f(1)=0 \quad f(0)=2 \tag{12}$$
$$f(2)=1 \quad f(1)=1 \quad f(0)=0 \tag{13}$$
$$f(2)=1 \quad f(1)=1 \quad f(0)=1 \tag{14}$$
$$f(2)=1 \quad f(1)=1 \quad f(0)=2 \tag{15}$$
$$f(2)=1 \quad f(1)=2 \quad f(0)=0 \tag{16}$$
$$f(2)=1 \quad f(1)=2 \quad f(0)=1 \tag{17}$$
$$f(2)=1 \quad f(1)=2 \quad f(0)=2 \tag{18}$$
$$f(2)=2 \quad f(1)=0 \quad f(0)=0 \tag{19}$$
$$f(2)=2 \quad f(1)=0 \quad f(0)=1 \tag{20}$$
$$f(2)=2 \quad f(1)=0 \quad f(0)=2 \tag{21}$$
$$f(2)=2 \quad f(1)=1 \quad f(0)=0 \tag{22}$$
$$f(2)=2 \quad f(1)=1 \quad f(0)=1 \tag{23}$$
$$f(2)=2 \quad f(1)=1 \quad f(0)=2 \tag{24}$$
$$f(2)=2 \quad f(1)=2 \quad f(0)=0 \tag{25}$$
$$f(2)=2 \quad f(1)=2 \quad f(0)=1 \tag{26}$$
$$f(2)=2 \quad f(1)=2 \quad f(0)=2 \tag{27}$$

All of them, in fact, have a polynomial representation. Letting f be any of the functions listed above where $f(0)=a$, $f(1)=b$, and $f(2)=c$, we can express $f(x)$ as:

$$f(x) = \frac{a}{2}(x-1)(x-2) - bx(x-2) + \frac{c}{2}x(x-1)$$

Since:

$$\begin{aligned}
f(0) &= \frac{a}{2}(0-1)(0-2) - b(0)(0-2) + \frac{c}{2}(0)(0-1) \\
&= \frac{a}{2}(-1)(-2) \\
&= \frac{a}{2}(2) \\
&= a
\end{aligned}$$

$$\begin{aligned}
f(1) &= \frac{a}{2}(1-1)(1-2) - b(1)(1-2) + \frac{c}{2}(1)(1-1) \\
&= -b(1)(-1) \\
&= -b(-1) \\
&= b
\end{aligned}$$

$$\begin{aligned}
f(2) &= \frac{a}{2}(2-1)(2-2) - b(2)(2-2) + \frac{c}{2}(2)(2-1) \\
&= \frac{c}{2}(2)(1) \\
&= \frac{c}{2}(2) \\
&= c
\end{aligned}$$

Thus, since such an f is clearly polynomial and can be used to construct all the functions listed above, all the functions in the vector space ${F_3}^{F_3}$ are polynomial.

(c) Give a linearly independent list of polynomial functions that spans $\mathcal{P}(F_3)$.

Solution:

We know that two polynomial functions $p, q \in \mathcal{P}(F_3)$ are equivalent if $\forall x \in F_3, p(x) = q(x)$. Therefore, we know that there are only 27 unique polynomial functions, since $P(F_3) = {F_3}^{F_3}$ as determined in the previous question.

Since every one of these polynomial functions p can be described by the three-tuple (a, b, c), where $p(0) = a, p(1) = b, p(2) = c$, and since we know that $(1, 0, 0), (0, 1, 0), (0, 0, 1)$ are a list of independent vectors in $\mathcal{P}(F_3)$ which span all such (a, b, c), they must all span $\mathcal{P}(F_3)$, and so the polynomial functions they correspond to must constitute a list of vectors which span $\mathcal{P}(F_3)$. Therefore, the polynomial functions:

$$
\begin{aligned}
p_1(x) &= \frac{1}{2}(x-1)(x-2) \\
p_2(x) &= -(x)(x-2) \\
p_3(x) &= \frac{1}{2}(x)(x-1)
\end{aligned}
$$

Constitute an independent list of vectors p_1, p_2, p_3 which span $\mathcal{P}(F_3)$.